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Vision-Language Agents Can Learn in Their Own Latent Worlds

Dwell Replaces Pixel-Space World Models with Recurrent Latent Transition in Frozen V-JEPA2 Embeddings, Cutting Imagination Compute by 2.3×

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Dwell builds a world model for vision-language agents that predicts the next latent state directly in the frozen embedding space of V-JEPA2, rather than reconstructing future frames. This removes the costly pixel decoder from model-based RL for VLMs, reducing imagination compute by 2.3× while achieving the strongest overall performance among trainable methods across five interactive visual environments.

AT A GLANCE

Problem: Pixel-space world models for VLM agents are computationally expensive and may focus on perceptually salient but task-irrelevant details.

Why it matters: Sample-efficient VLM agents are needed for tasks where environment interaction is costly (robotics, real-world planning, interactive apps).

Method: Recurrent latent transition model in frozen V-JEPA2 embedding space; VLM imagines ahead in latent space; no pixel decoder.

Result: Best overall performance across 5 visual environments among trainable methods; 2.3× lower imagination compute.

Evidence: 5 interactive visual environments; ablation over world model type and encoder choice.

The Big Picture

Model-based reinforcement learning (MBRL) aims to improve sample efficiency by training a world model—a learned simulator of the environment—and using it to plan or generate synthetic experience for policy learning. In the RL literature, Dreamer and its successors demonstrated that pixel-space world models trained with reconstruction objectives could dramatically outperform model-free methods on visual control tasks.

For vision-language agents, however, pixel-space world models introduce a double burden. First, they require a large image decoder to reconstruct future frames, which is computationally expensive. Second, the reconstruction objective encourages the world model to represent all visual details, including those that are irrelevant to the agent's task (e.g., the exact texture of a wall, the shadow of an object). This can distract representation learning from task- relevant features.

Dwell (Dreaming With Latent Embeddings for VLMs) proposes a minimal world model that operates entirely in latent space, predicting the next latent state rather than the next frame. The key design choice is to ground the latent space in a frozen pretrained visual encoder (V-JEPA2) rather than learning it jointly with the world model.

Why Existing Approaches Fall Short

Pixel-space Dreamer variants: These reconstruct future frames using a stochastic state space model and a convolutional decoder. For VLM agents, the decoder must handle high-resolution images aligned with the language context, which is expensive. The reconstruction loss does not distinguish task-relevant from task-irrelevant visual details.

End-to-end latent world models: Training the latent representation jointly with the world model is unstable for vision-language agents, where the latent space must simultaneously support language-conditioned reasoning and state prediction. Prior work (TD-MPC2 for VLMs) addresses this with complex auxiliary objectives; Dwell eliminates the problem by using a frozen encoder.

Model-free fine-tuning: Direct RL fine-tuning of VLMs without a world model is sample-inefficient and can catastrophically forget pretrained language abilities if the RL signal is too strong.

Core Mechanism

Frozen V-JEPA2 Encoder. All visual observations are encoded by a V-JEPA2 ViT-L encoder whose weights are frozen throughout training. V-JEPA2 was pretrained on video prediction; its embeddings are therefore already rich in temporal and semantic information, well-suited for predicting the next state.

Recurrent Latent Transition Model (RLTM). The core world model is a recurrent neural network: $h_{t+1} = \text{RLTM}(h_t, z_t, a_t, g_t)$ where h_t is the recurrent state, $z_t = \text{V-JEPA2}(o_t)$ is the current visual embedding, a_t is the action embedding, and g_t is the language goal embedding from the VLM.

Imagination and Policy Learning. For policy updates, the agent unrolls the RLTM for H steps in latent space: starting from a real observed state z_t, it generates an

imagined trajectory $(\hat{z}_{t+1}, \hat{z}_{t+2}, \ldots, \hat{z}_{t+H})$. A value head on top of the RLTM outputs reward predictions for each imagined state. The policy is updated by actor-critic on the imagined trajectories.

Technical Formulation

Let $z_t = f_{\text{JEPA}}(o_t) \in \mathbb{R}^d$ be the frozen V-JEPA2 embedding and $g = f_{\text{VLM}}(\ell)$ the language goal encoding. The RLTM is trained by minimizing:

$$\mathcal{L}_{\text{RLTM}} = \sum_{t=1}^{T} \|\hat{z}_t - z_t\|^2 + \lambda_r \sum_{t=1}^{T} (\hat{r}_t - r_t)^2$$

where $\hat{z}_t = h_t W_z$ and $\hat{r}_t = h_t w_r$ are predicted from the recurrent state.

The policy $\pi_\theta(a \mid h, g)$ and value $V_\phi(h)$ are trained with a model-based actor-critic objective over imagined trajectories of length H:

$$\mathcal{L}_{\text{actor}} = -\mathbb{E}_{\hat{\tau}} \left[\sum_{k=0}^{H} \gamma^k \hat{r}_{t+k} + \lambda_e \mathcal{H}[\pi_\theta] \right]$$

$$\mathcal{L}_{\text{critic}} = \mathbb{E}_{\hat{\tau}} \left[\left(V_\phi(h_{t+k}) - \hat{V}_{\text{target}} \right)^2 \right]$$

Learning or Inference Procedure

Warm-up: Collect N_{warmup} real environment transitions to initialize the replay buffer.

Alternate:

1. Collect M real steps with the current policy.
2. Update RLTM on the replay buffer for K gradient steps.
3. Unroll H-step imagined trajectories from the buffer.
4. Update π_θ and V_ϕ on imagined + real data.

The VLM backbone is fine-tuned with a small learning rate to adapt language-goal encoding to the new tasks, but the V-JEPA2 encoder is never updated.

What the Guarantee Says

No formal guarantee is claimed. The paper argues informally that using a pretrained visual encoder that already captures temporal dynamics (V-JEPA2) as the latent space provides a better-conditioned prediction target than a randomly initialized latent, reducing the risk of the world model diverging from the true dynamics. Empirically, the RLTM prediction error is lower for V-JEPA2 than for randomly initialized or reconstructed latents, supporting this claim.

Experimental Findings

- **Best overall:** Dwell achieves the highest mean task success across 5 visual environments among all trainable methods.
- **Compute reduction:** Imagination using the RLTM requires **2.3× less compute** than a pixel-space Dreamer baseline with the same planning horizon H.
- **Sparse reward advantage:** On tasks with sparse rewards, Dwell's margin over model-free baselines is 12.4 percentage points; on dense-reward tasks, the margin narrows to 3.1 points.
- **V-JEPA2 vs. other encoders:** Using CLIP instead of V-JEPA2 as the frozen encoder reduces performance by 8.2% on average, confirming the value of a temporally-aware encoder for the world model.

Ablations and Interpretation

Latent prediction vs. pixel reconstruction: Replacing RLTM with a pixel-decoder world model of similar capacity reduces performance by 4.1% and increases imagination compute by 2.3×, validating the latent-only design.

Frozen vs. fine-tuned V-JEPA2: Fine-tuning V-JEPA2 alongside the world model degrades performance by 5.3%, likely due to the joint instability of learning the latent space and predicting in it simultaneously.

Imagination horizon H: Performance peaks at $H = 5$ steps and degrades for $H > 10$, consistent with the compounding prediction error in long latent rollouts.

RLTM size: Scaling the RLTM from a 2-layer GRU to a 4-layer Transformer adds 2.7% success, suggesting that the world model's capacity is not yet a bottleneck at the evaluated environment scales.

Reference

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